

SPREAD SPECTRUM SIGNALS FOR DIGITAL COMMUNICATION

As we studied number of digital modulation technique which are more concerned about the efficient utilization of bandwidth and transmitting power, these two are major resources of any communication system. There are some other communications required, where it is necessary for the system to resist external interference and to make it difficult for unauthorized receivers to receive the message being transmitted. This type of communication is called "secured communication." The modulation technique used for this secured communication is spread spectrum modulation technique, which is basically implemented for military applications. The interference in the transmission channel may be unintentional, because the other user may be transmitting through same channel.

The spread spectrum modulation can be defined in two parts are

- (i) The transmitted data sequence occupies much more bandwidth than the minimum bandwidth required for transmission of actual data sequence.
- (ii) The transmitting signal spectrum is spreaded at the transmitter and despreaded at the receiver is obtained "special code" which is independent of the data sequence.

The first part of definition is satisfied by other modulation techniques like frequency modulation, PCM etc, but they do not satisfy second part.

DC
A SEM
MMN SIG

MODEL OF SPREAD SPECTRUM DIGITAL COMMUNICATION

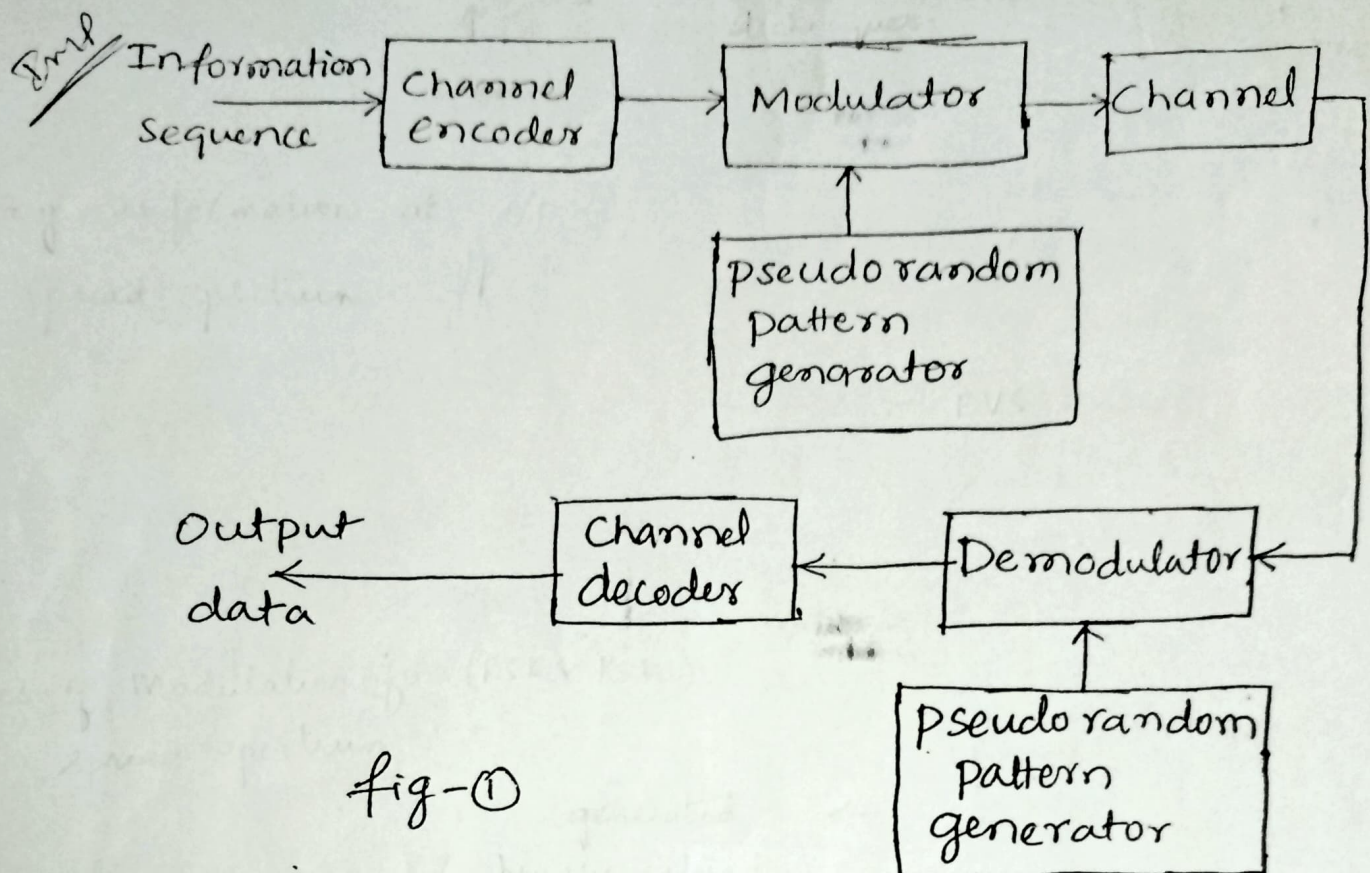


fig-①

The block diagram of Spread Spectrum digital communication system is shown in fig①, in which binary information as its input at transmitter and receiver outputs. The channel encoder encodes the input binary sequence according to some error control technique but the channel decoder performs the reverse function of channel encoder and gives the original binary sequence as its output.

Basically we use two types of modulation used for spread spectrum modulation are PSK and FSK.

PSK is used where phase coherence is maintained between transmitter and receiver. FSK modulation is used where phase coherence can not be maintained due to time invariant effect of communication link.

From the block diagram shown, we have used two identical pseudo random pattern generators, one at the transmitter and other at the receiver. PN Sequence generator generates binary valued sequence which is impressed on the transmitted signal at the modulator and removed from the received signal at the receiver. The two PN sequence generators are synchronized between transmitter and receiver.

While transmitting the information bearing signal through transmission channel there may be a chance of introducing interference. This interference signal is categorized at the receiver, whether it is broadband or narrow band relative to bandwidth of information bearing signal. at Receiver $\rightarrow$ the interference signal is categorized

The PN sequence generated at the modulator is used in conjunction with the PSK modulation to shift the phase of the PSK signal resulting signal is called Direct Sequence (DS) or PN spread spectrum signal.

When PN sequence generated at the modulator is used in conjunction with M-ary FSK resulting signal is called Frequency hopped (FH) spread spectrum signal.

APPLICATIONS OF SPREAD SPECTRUM MODULATION

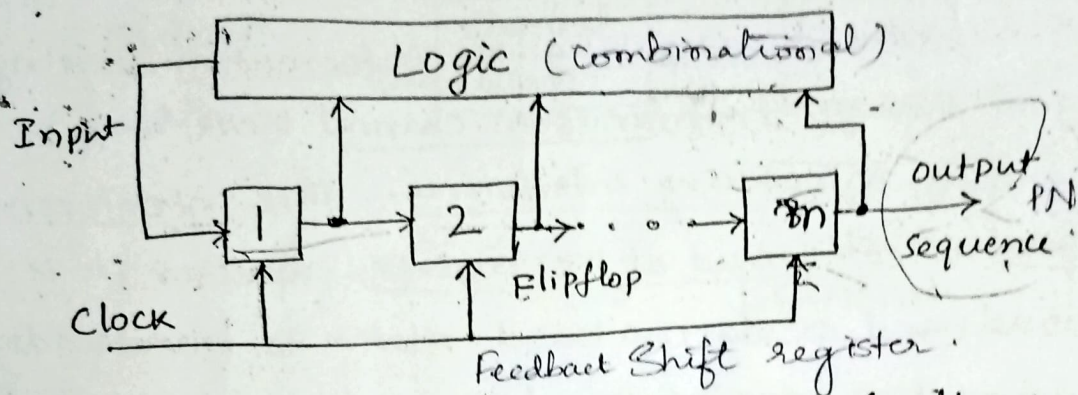
1. Antijam capability
2. Interference rejection
3. Multiple access capability
4. Multipath protection
5. Low probability of Intercept (LPI)

6. Secured communication

7. Improved spectral efficiency

GENERATION OF PN SEQUENCES

The pseudo noise or pseudo random sequence is noise like high frequency signal. The sequence is not completely random instead it is generated by a well defined function or logic



The PN sequence can be generated by a feedback shift register and a combinational logic

⊗ The shift register consists of 'n' flip-flops. The data of one flip-flop is given to next flip-flop whenever clock pulse is applied

⊗ The outputs of flip-flops are given as an input to combinational logic circuit.

⊗ The output of combinational logic is given as an input for first flip-flop in a shift register

⊗ The PN sequence is generated at the output from last flip-flop

⊗ The PN sequence generated at the output of flip-flop is repeated after 2^n digits. Hence also called period of output sequence.

(*) Normally the logic circuit is mod-2 adders.

If shift register enters in Zero state, it will not come out of it and output sequence will be zero only. To prevent this, the Zero state of the shift register is not allowed. Therefore total number of states in m-stage shift register will be $2^m - 1$. Hence PN Sequence is also called maximum length sequence.

(*) the number of stages of shift register and period of maximum length sequence is tabulated below.

m	$L = 2^m - 1$
3	7
4	15
⋮	⋮
15	32767

→ properties are applied to periodic sequence to test for appearance of Randomness

Properties of Maximum length Sequence

There are three properties that can be applied to any periodic sequence to test for appearance of randomness

Property 1:- In each period of maximum length sequence the number of 1's should always be one greater than number of 0's, this property is called "balance property."

the balance property ensures that the probability of Symbol '0' and Symbol '1' are equal.

Property 2:- A run is defined as a sequence of similar binary digits (0's or 1's). the length of the run is equals to the number of similar digits in the run. In every period one-half the runs of each type (1's or 0's) are of the length 1; about one-fourth of the run are of length 2, one-eighth of run are of length 3 and so on. But the number of runs in a given period is $\frac{(L+1)}{2}$, where $L = 2^m - 1$

Ex:- ① 0011101
 $R_1 \quad R_2 \quad R_3 \quad R_4$

$$\text{The number of runs} = \frac{7+1}{2} = 4$$

$$\frac{1}{2} \times 4 = 2 \text{ of length 1}$$

$$\frac{4}{4} = 1 \text{ of length 2}$$

$$\frac{1}{8} \times 4 = \text{Can not be find } \text{Since sequence length is not sufficient.}$$

Ex② 111100010011010
 $R_1 \quad R_2 \quad R_3 \quad R_4 \quad R_5 \quad R_6 \quad R_7 \quad R_8$

$$\text{the number of runs} = \frac{15+1}{2} = 8$$

$$\frac{1}{2} \times 8 = 4 \quad \text{of length 1}$$

$$\frac{1}{4} \times 8 = 2 \quad \text{of length 2}$$

$$\frac{1}{8} \times 8 = 1 \quad \text{of length 3}$$

$$\frac{1}{16} \times 8 = \text{Can not be found since length of sequence is not sufficient.}$$

Property 3:- Let $\{C_0, C_1, \dots, C_{L-1}\}$ be the maximum length sequence with a period of $L = 2^m - 1$ generated by 'm' stage linear shift register. Then cyclic autocorrelation function of the sequence is defined as

$$R_c(\tau) = \frac{1}{L} \sum_{i=0}^{L-1} C_i C_{(i \oplus \tau) \bmod L} \quad \tau = 0, 1, 2, \dots, L-1$$

where $\oplus$ is the shift

Further an autocorrelation function of maximum length sequence is also periodic with the same period 'L'

Thus

$$R_c(\tau) = \begin{cases} 1 & \text{for } \tau = kL, k = 0, 1, 2, \dots \\ -\frac{1}{L} & \text{for } \tau \neq kL, k = 0, 1, 2, \dots \end{cases}$$

Ex: 0010111

$$\{C_0, C_1, C_2, C_3, C_4, C_5, C_6\} = \{0010111\} = \{-1, -1, 1, -1, 1, 1, 1\}$$

$$\therefore R(z) = \frac{1}{7} \sum_{i=0}^6 C_i (i \oplus z) \bmod 7$$

$$\underline{R(0)}$$

$$\cancel{C_i = 0}$$

$$C_i = \begin{matrix} -1 & -1 & 1 & -1 & 1 & 1 & 1 \end{matrix}$$

$$C(i \oplus 0) = \begin{matrix} -1 & -1 & 1 & -1 & 1 & 1 & 1 \end{matrix}$$

$$\begin{matrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \end{matrix}$$

$$R(0) = \frac{1}{7} (C_0^2 + C_1^2 + C_2^2 + C_3^2 + C_4^2 + C_5^2 + C_6^2)$$

$$= \frac{7}{7} = 1$$

To compute $R(1)$

$$C_i = \begin{matrix} C_0 & C_1 & C_2 & C_3 & C_4 & C_5 & C_6 \end{matrix}$$

$$C(i \oplus 1) = \begin{matrix} C_1 & C_2 & C_3 & C_4 & C_5 & C_6 & C_0 \end{matrix}$$

$$R(1) = \frac{1}{7} (C_0 C_1 + C_1 C_2 + C_2 C_3 + C_3 C_4 + C_4 C_5 + C_5 C_6 + C_6 C_0)$$

$$= \frac{1}{7} (1 - 1 - 1 - 1 + 1 + 1 - 1)$$

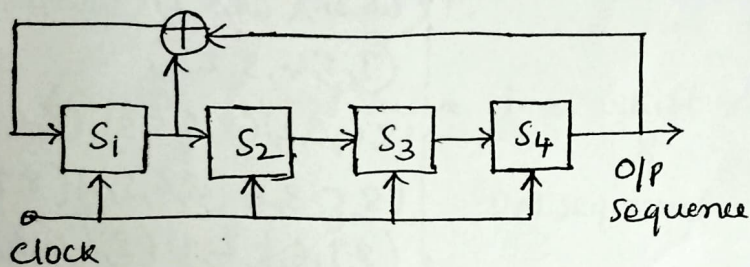
$$= \frac{-1}{7}$$

$$R(2) = R(3) = R(4) = R(5) = R(6) = -\frac{1}{7}$$

$$\therefore R(z) = \begin{cases} 1 & \text{for } z = 0, \pm 7, \dots \\ -\frac{1}{7} & \text{for } z \neq 0, \pm 7, \dots \end{cases}$$

Ex 1:- A PN Sequence is generated using feedback Shift register of length Four. Find the generated output sequence if the initial contents of the shift registers are 1000. If the chip rate is 10^7 chips/sec, calculate the chip, PN sequence duration and period of output sequence. Draw its Schematic arrangement.

Soln:- (*) There are Four stages of Shift registers. To generate maximum length sequence, the output of 4th & 1st flipflop are given as input to Ex-OR Logic



(*) The generated PN sequence as

000111101011001
 $\underbrace{\quad}_{R_1} \quad \underbrace{\quad}_{R_2} \quad \underbrace{\quad}_{R_3} \quad \underbrace{\quad}_{R_4} \quad \underbrace{\quad}_{R_5} \quad \underbrace{\quad}_{R_6} \quad \underbrace{\quad}_{R_7} \quad \underbrace{\quad}_{R_8}$

(*) The chip rate $R_c = 10^7$ chips/sec

$$\therefore \text{chip duration} = \frac{1}{R_c} = \frac{1}{10^7} = 0.1 \mu\text{sec}$$

(*) To obtain length of PN sequence

$$L = 2^m - 1$$

$$= 2^4 - 1 = 15$$

(*) To obtain period of output sequence is

$$T_b = N T_c \text{ or } L T_c$$

$$= 15 \times 0.1 \mu\text{sec}$$

$$= \underline{1.5 \mu\text{sec}}$$

S_1	S_2	S_3	S_4	PN Seq.
1	0	0	0	0
1	1	0	0	0
1	1	1	0	0
1	1	1	1	1
0	1	1	1	1
1	0	1	1	1
0	1	0	1	1
1	0	1	0	0
1	1	0	1	1
0	1	1	0	0
0	0	1	1	1
1	0	0	1	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	1
1	0	0	0	0

Selection of maximum length sequence

number of stages 'm'	Feedback taps
2	(2, 1)
3	(3, 1)
4	(4, 1)
5	(5, 2) (5, 4, 3, 2) (5, 4, 2, 1)
6	(6, 1) (6, 5, 2, 1) (6, 5, 3, 2)
7	(7, 1) (7, 3) (7, 3, 2, 1) (7, 4, 3, 2) (7, 6, 4, 2) (7, 6, 3, 1) (7, 6, 5, 2) (7, 6, 5, 4, 2, 1) (7, 5, 4, 3, 2, 1)
8	(8, 4, 3, 2) (8, 6, 5, 3) (8, 6, 5, 2) (8, 5, 3, 1) (8, 6, 5, 1) (8, 7, 6, 1) (8, 7, 6, 5, 2, 1) (8, 6, 4, 3, 2, 1)

Ex②: A PN Sequence is generated using linear feed-back shift register with number of stages equals to 10. The chip rate is $10^7/\text{sec}$. Find the following.

- PN Sequence length
- chip duration of PN Sequence
- Period of PN Sequence

Soln.

- PN Sequence length is $2^{10} - 1 = 1023$
- chip duration $T_c = \frac{1}{R_c} = \frac{1}{10^7} = 0.1 \mu\text{sec}$
- Period of PN Sequence is

$$T = 1023 \times 0.1 \mu\text{sec} \\ = \underline{\underline{102.3 \mu\text{sec}}}$$

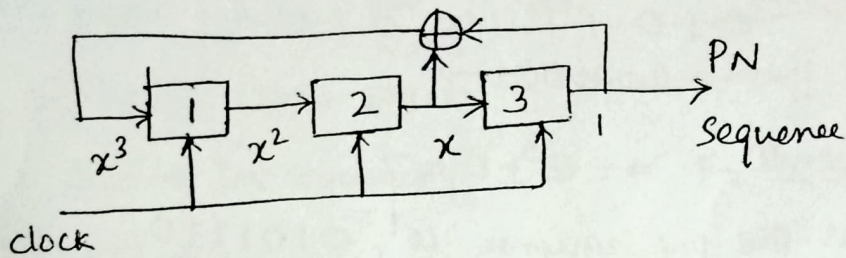
Ex (3): Generate a PN sequence for the polynomial $f(x) = x^3 + x + 1$ and check the properties of PN sequence

Soln :- $f(x) = x^3 + x + 1$

Considering R.H.S and equate to zero

$$x^3 + x + 1 = 0$$

$$x^3 = x + 1 \quad \{ \text{neglect the sign} \}$$



Assume initial value in a shift register is 111

S_1	S_2	S_3	PN sequence
1	1	1	1
0	1	1	1
0	0	1	1
1	0	0	0
0	1	0	0
1	0	1	1
1	1	0	0
1	1	1	1

↓ Sequence repeats

$$L = 2^m - 1 = 2^3 - 1 = 7$$

∴ The Sequence is 1110010 [111-1-1-1]

Check the three properties

Ex(4) :- A 3-stage shift register with a linear feedback generates the sequence

0 1 0 1 1 1 0 0 1 0 1 1 1 0 ...

- Determine the period of given infinite sequence
- Verify the three properties

Soln

a) The given sequence is

0 1 0 1 1 1 0 0 1 0 1 1 1 0 ...
| ← 1-period → |

$$L = 2^m - 1 = 2^3 - 1 = 7$$

∴ The PN sequence is 0101110

b) ~~⊗~~ Balance property

no of 1's is equals to 4

no of 0's is equals to 3

Hence satisfied

~~⊗~~ Run property

0 1 0 1 1 1 0
└─┘ └─┘ └─┘ └─┘
R₁ R₂ R₃ R₄ R₅

But number of runs = $\frac{L+1}{2} = \frac{7+1}{2} = 4$

Hence Run property is not satisfied. If we make a circular shift of one time then the sequence becomes valid

0 1 0 1 1 1 0 1 time circular shift

0 0 1 0 1 1 1
└─┘ └─┘ └─┘ └─┘
R₁ R₂ R₃ R₄

$$\frac{1}{2} \times 4 = 2 \quad \text{of one bit}$$

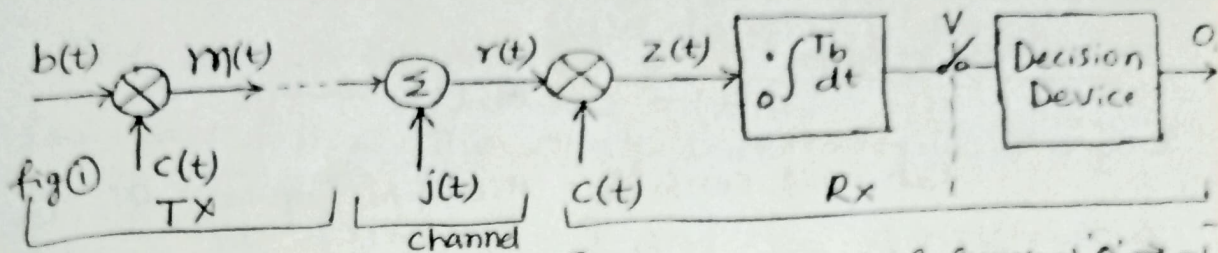
$$\frac{1}{4} \times 4 = 1 \quad \text{of two bits}$$

Hence Run property is satisfied.

(*) Verify the Autocorrelation property

DIRECT SEQUENCE SPREAD SPECTRUM SIGNALS [DS-SS]

(*) DS-SS Base band transmission



$b(t) \Rightarrow$ Data Signal [Symbol 1 $\rightarrow +1$ & Symbol 0 $\rightarrow -1$]

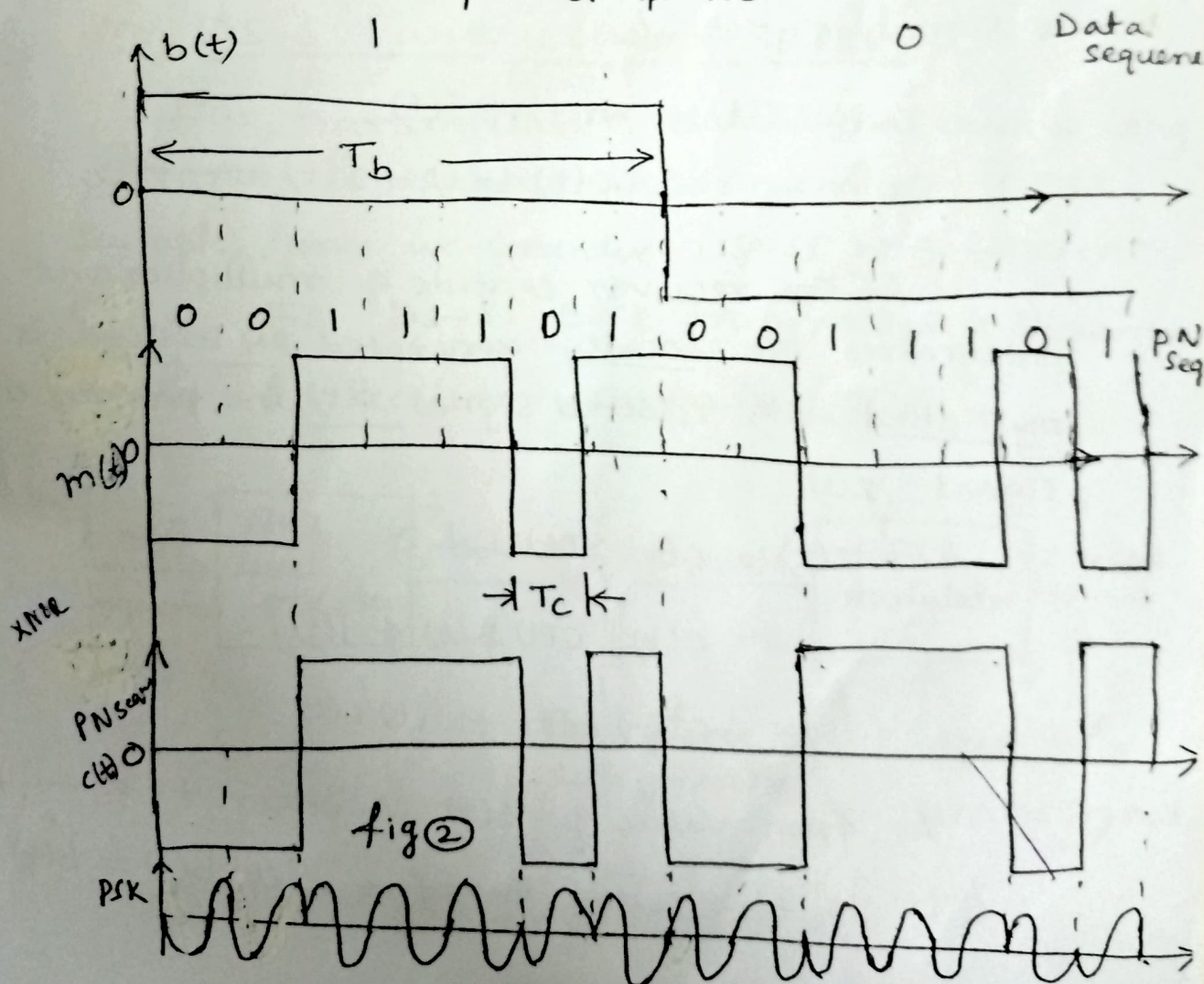
$c(t) \Rightarrow$ PN Sequence [1 $\rightarrow +1$ & 0 $\rightarrow -1$]

$j(t) \Rightarrow$ Interference signal

$z(t) \Rightarrow$ output of multiplier at Rx

$m(t) \Rightarrow$ o/p of multiplier at Tx

$V \Rightarrow$ Sampled amplitude



The data signal $b(t)$ and PN sequence signal $c(t)$ are applied to a product modulator. The output of a modulator is a "wide spectrum signal". The bandwidth of signal $m(t)$ is very large compare to bandwidth of signal $b(t)$

$$m(t) = c(t) \otimes b(t) \text{ --- (1)}$$

Let us consider the PN Sequence as 0011101

The fig (2) shows the waveforms of ~~m~~ $b(t)$, $c(t)$ and $m(t)$. Here the signal $m(t)$ is directly transmitting through the channel ~~at~~ without any basic modulation, hence it is called base band transmission.

During the transmission of signal $m(t)$, the noise $j(t)$ gets interfered with message signal and resulting signal is $x(t)$ i.e

$$\begin{aligned} x(t) &= m(t) + j(t) \\ &= c(t) b(t) + j(t) \text{ --- (2)} \end{aligned}$$

At the receiver consists of multiplier and integrator. The locally generated PN sequence is multiplied with received signal $x(t)$ and produces a signal $z(t)$

$$\begin{aligned} z(t) &= c(t) x(t) \text{ --- (3)} \\ &= c(t) [c(t) b(t) + j(t)] \\ &= c^2(t) b(t) + j(t) c(t) \end{aligned}$$

As from value of $c(t)$ is either +1 or -1, then $c^2(t)$ is always +1. Therefore $c^2(t) b(t) = b(t)$

$$\therefore z(t) = b(t) + c(t) \cdot j(t) \quad \text{--- (4)}$$

The $c(t)$ is a wideband signal and the product of $c(t) \cdot j(t)$ is also a wideband signal, but output of multiplier is integrated over T_b then it gives only the first term in equation (4) i.e. integrator acts as a low-pass filter. The o/p of Integrator is sampled at $t = nT_b$ [$n=1, 2, \dots$] and amplitude of samples are compared at decision device with its threshold and gives desired output data

$$\begin{aligned} \text{O/p} &= 1 & V > 0 \\ &= 0 & V < 0 \end{aligned}$$

* (*) DS-SS with coherent BPSK.

For transmission of data signal over a long distance DS-SS baseband transmission is not suitable, hence we consider DS-SS with coherent BPSK. The DS-SS BPSK transmitter is shown in fig ① & Receiver is shown in fig ②

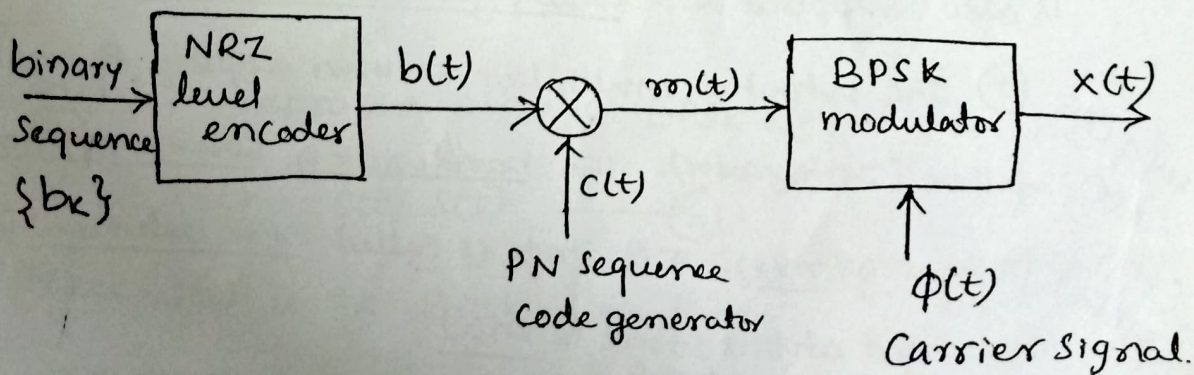


fig-①

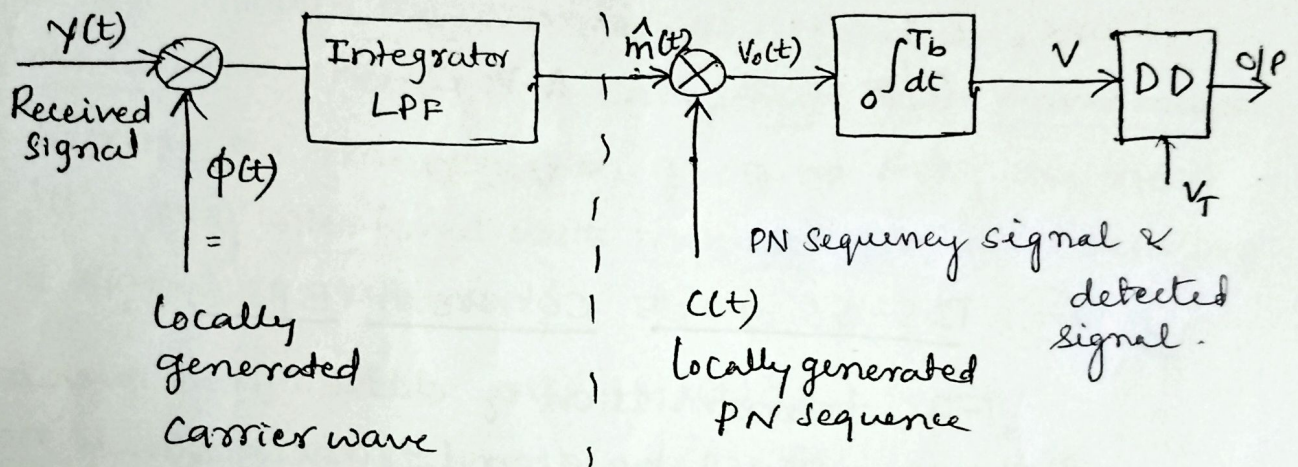
In this system, uses two stages of modulation
 & $b(t)$ is multiplied with $c(t)$ and $m(t)$ is PSK
 modulated with sinusoidal carrier wave $\phi(t)$.

$$\phi(t) = \sqrt{2P_c} \sin(2\pi f_c t) \quad \text{--- ①}$$

then transmitted signal is

$$x(t) = \sqrt{2P_c} m(t) \sin(2\pi f_c t) \quad \text{--- ②}$$

When $m(t) = 1$ (+1) 0° phase shift
 $= 0$ (-1) 180° phase shift



← Coherent BPSK → | ← Despreading of DS-SSS →

(*) The received signal $y(t)$ is applied to the multiplier which is also supplied with locally generated coherent carrier.

BW of LPF = $m(t)$ (*) The output of multiplier is then applied to the low-pass filter (Integrator). the bandwidth of Low-pass filter is equal to that of $m(t)$. this stage is called BPSK detector. The signal obtained at this stage is $m(t)$

(*) This signal is supplied to the second demodulator which despreads the signal. The local PN sequence is exactly

replica of that used in the transmitter

(*) The integrator integrates the products of detected message signal and PN signal over one bit period ' T_b ' $\boxed{\int_0^{T_b} dt}$

(*) The decision is then taken depending upon the polarity of output ϕ of the integrator

FREQUENCY HOP SPREAD SPECTRUM (FH-SS)

Frequency hopping means transmitting data bits in different frequency slots or hops. The total bandwidth required for transmitting frequency hopped signal is equal to sum of all frequency slots or hops.

Since frequency hopping is random, i.e. known only to the recognized transmitter and receiver, hence unauthorized receivers find it difficult to receive frequency hopped signal. It is because transmitting signal occupies in the bandwidth in terms of GHz, which is much greater than the bandwidth required for transmitting actual signal.

In frequency hop spread spectrum, the carrier frequency hop is varied randomly from one frequency to another. Normally M-ary FSK modulation is used along with frequency-hop spread spectrum. Basically there are two types of Frequency hopping spread spectrums, which are

1. Slow frequency hop - spread spectrum [SH-SS]
2. Fast frequency hop - spread spectrum [FH-SS]

When 'M' symbols [$M=2^k$] are transmitted with the help of M-ary FSK modulation system, the spread spectrum modulation is to be used to generate

wide band signal. The spread spectrum carrier frequency hops are random for one frequency/symbol to another frequency/symbol.

The rate of change of frequency hops is called hop rate (R_h) and the rate at which K -bits symbol of input data sequence is generated is called Symbol rate (R_s).

$\Rightarrow$ 1. Slow frequency Hopping (SH-SS)

When several symbols of given binary sequence are transmitted in one frequency hop is called slow frequency hopping.

Normally ' K ' successive bits of input data sequence represents $2^K = M$ symbols. The ' M ' symbols are transmitted with the help of M -ary FSK modulation system. When spread spectrum modulation is used, then M -ary FSK signal is further modulated to wide band PN sequence. The spread spectrum carrier frequency hops randomly from one frequency to another. In slow frequency hopping multiple symbols are transmitted in one frequency hop.

The block diagram of slow-frequency hopping transmitter and receiver is shown in fig-① and fig-②.

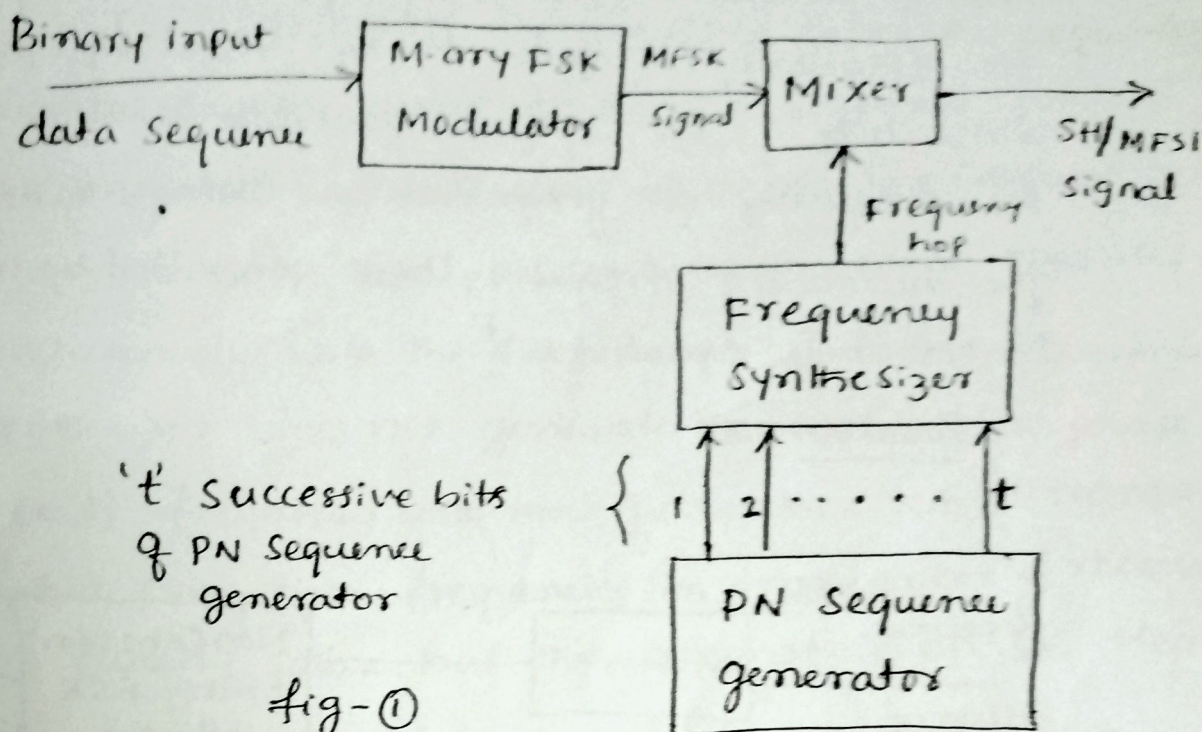


fig-①

The input binary data sequence is applied to the M-ary FSK modulator. The output of modulator is a particular frequency depending upon input symbol and its output is given to mixer. The another input sig. given to the mixer is again a particular frequency from frequency synthesizer. The output of frequency synthesizer at particular instant is frequency slot or hop. This hop is mixed with FSK signal. The mixer circuit is designed to provide 'sum' frequency component of FSK signal and frequency hop from frequency synthesizer. The output of mixer is called FH/MFSK signal and is transmitted over a wide band channel.

The frequency hop produced by frequency

Synthesizer is controlled by PN Sequence generator.
 The 't' successive bits of PN sequence generator controls
 the frequency hop generated by frequency synthesizer.
 Since bits of PN sequence generator changes randomly
 the frequency hops generation also changes randomly.
 The number of frequency hops generated by frequency
 synthesizer is equal to 2^t .

Receiver

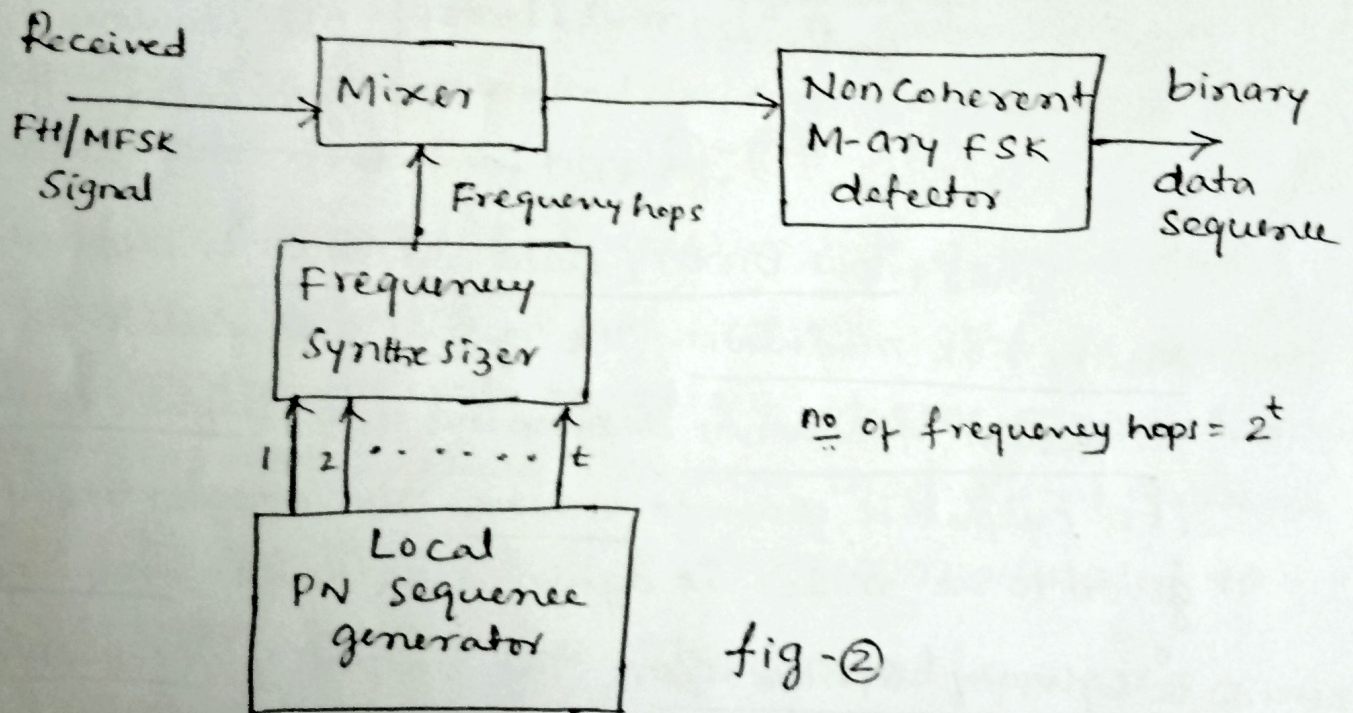


fig - ②

The block diagram shown in fig - ② is FH/MFSK receiver. The received FH/MFSK signal is applied to the mixer. The output of frequency synthesizer is also given to the mixer. At the receiver side the mixer circuit is designed to provide only the difference component. The output of mixer is given to the non-coherent M-ary FSK detector. The detector

detects particular symbol transmitted.

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In FH/MFSK system, the local PN sequence generator exactly generates the P.N sequence which was generated in the receiver at particular instant of time. That is the local PN sequence generator operates synchronously with the transmitter. But the frequency synthesizer does not generate the 'inphase' (or phase locked) frequency with that of synthesizer at transmitter. That is synthesizer frequencies in transmitter & receiver are exactly same but they are not in phase. Hence coherent detection is not possible in FH/MFSK receiver.

⇒ Chip rate for slow frequency hopping

In FH/MFSK the individual frequency of smallest duration is called chip. In slow frequency hopping multiple symbols are transmitted per hop. The chip rate R_c is equal to symbol rate R_s for slow frequency hopping.

$$R_c = R_s$$

$$R_c = \frac{R_b}{K}$$

where 'K' is number of bits/symbol

⇒ processing gain (PG)

The processing gain is defined as

$$PG = \frac{\text{Bandwidth of spreaded signal}}{\text{Bandwidth of unspreaded signal}}$$

Let f_s is the symbol frequency and requires 't' bits of PN sequence to represent each symbol. Therefore bandwidth of spreaded signal is $2^t f_s$. The bandwidth of unspreaded signal is equals to f_s . Hence processing gain will be

$$PG = \frac{2^t f_s}{f_s} = 2^t$$

Let us consider $t = 3$. ($2^3 = 8$), then we can generate 8- frequency hops over complete FH bandwidth, and assume that two binary data bits required to represent each symbol. The fig-(3) shows four possible symbols are transmitted in a single frequency hop. As shown in fig-(4) the two symbols occupy any two frequencies in one hop out of four. The chip rate is equal to symbol rate. The symbol 01 has FSK frequency of f_2 . And because of frequency hopping this frequency is increased to $f_H + f_2$. Similarly symbol 11 has frequency of $f_H + f_H$. The hop frequency f_H is

is controlled by PN sequence bit

2

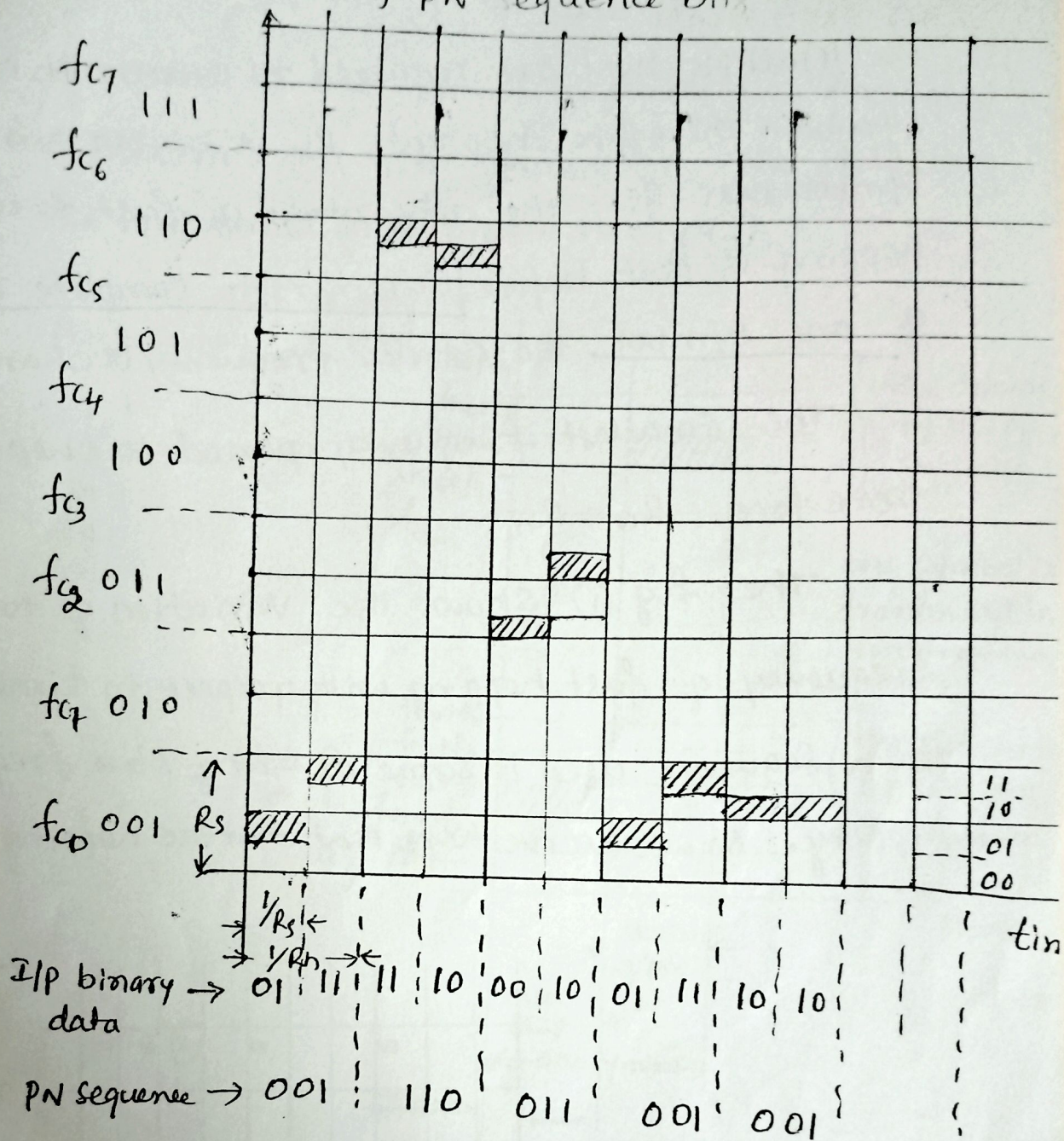


fig-③

4-ary FSK
8 $\rightarrow$ frequency hops

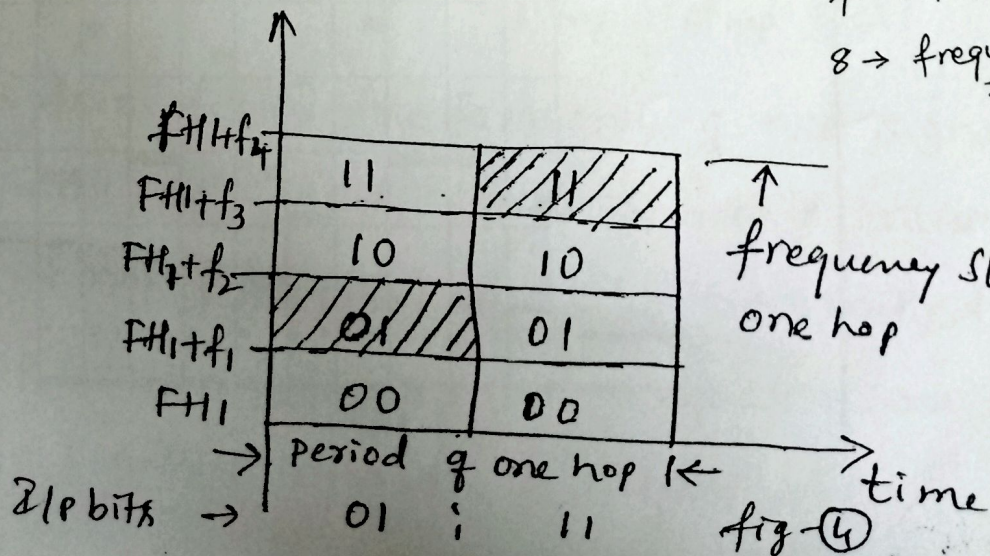


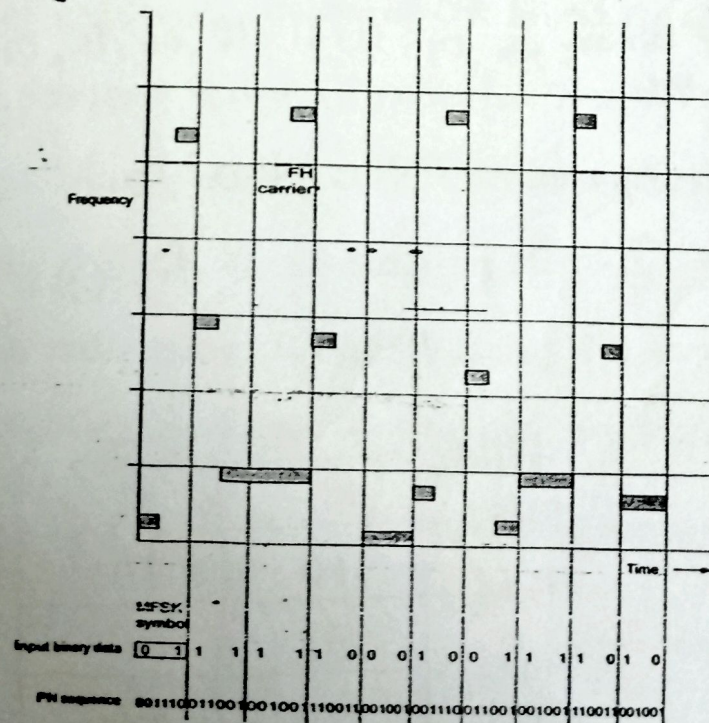
fig-④

⇒ Fast frequency hopping (FH-S)

Multiple hops are required to transmit one data symbol. Therefore hop rate R_h is higher than the symbol rate R_s . The advantage of fast frequency hopping is that before Jammers to complete reception of one symbol, the carrier frequency is changed.

The smallest frequency period is chip rate therefore $R_c = R_h$

The fig ① shows the variation of transmitted frequency of fast hopping with respect to time. The data sequence used is same as that of slow frequency hopping. Now observe here that ~~that~~ the hopping rate



From fig (1), the first two data bits '01' corresponds to one symbol requires two frequency hops to transmit this first symbol '01'. The details of this transmission is shown in fig-(2)

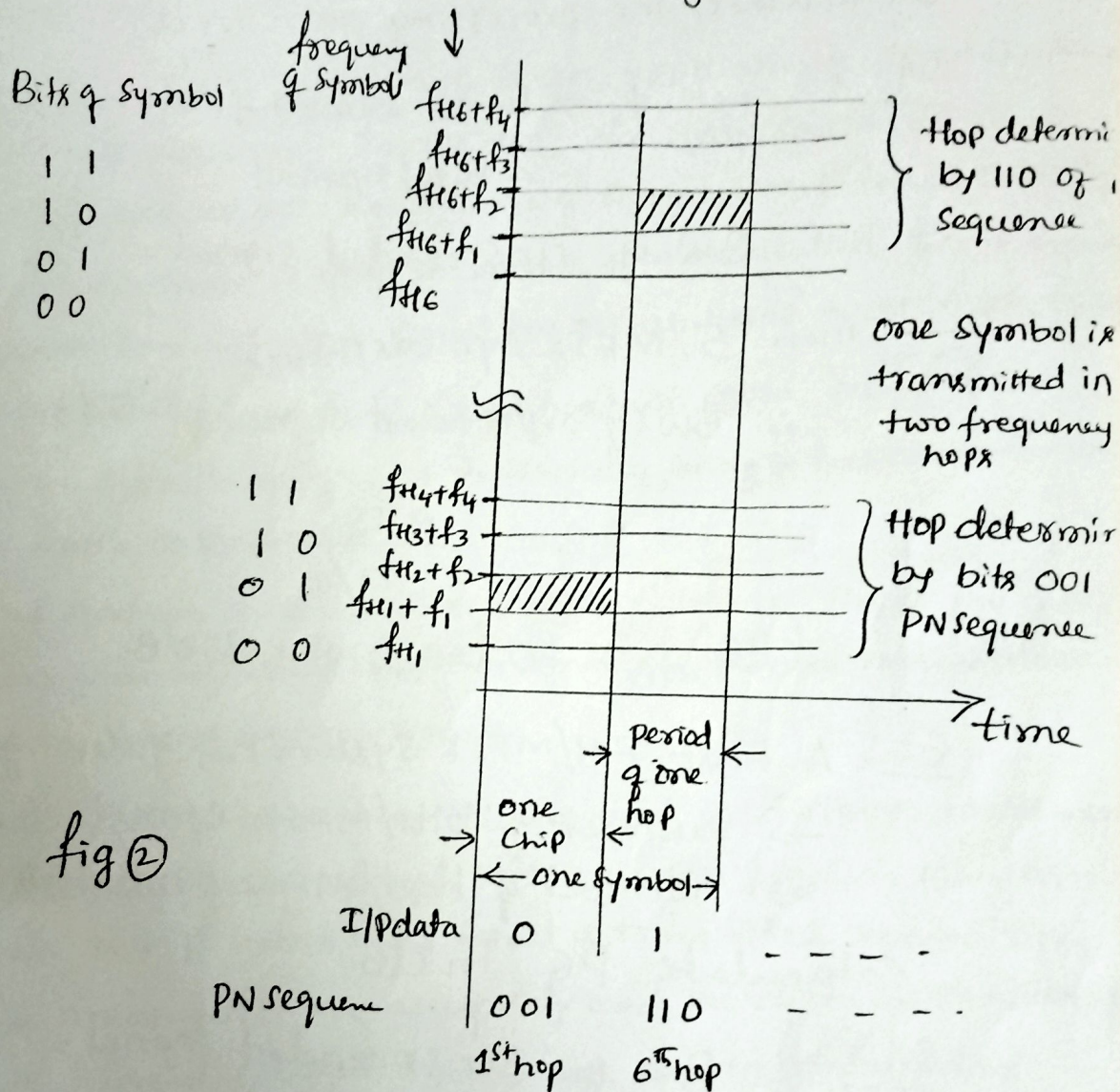


fig (2)

As from fig-(2) the frequency of FSK signal for symbol '01' is f_2 . This symbol is transmitted in first hop ($f_{H1} + f_2$) and also transmitted in sixth hop ($f_{H6} + f_2$). One chip is equal to one hop, it is the smallest interval in FH-SS.

Ex A slow FH/MFSK system has the following parameters

→ The number of bits/MFSK symbol = 4

→ The number of MFSK symbol/hop = 5

calculate the processing gain in dB

Soln. : Let $f_s \Rightarrow$ be the symbol frequency i.e. b/s
4 bits/symbol

$$\therefore \text{BW of unspread signal} = f_s/4$$

Then 5 MFSK symbols/hop

$$\therefore \text{BW of spread signal} = 5f_s$$

$$\therefore PG = \frac{5f_s}{f_s/4} = 20$$

$$(PG)_{dB} = 10 \log_{10}(20) = 13 \text{ dB.}$$

Ex A Fast FH/MFSK system has following parameters

→ number of bits/MFSK symbol = 4

→ number of hops/MFSK symbol = 4

calculate PG in dB

Soln. : BW of unspread signal = $f_s/4$

$$\text{BW of spread signal} = 4f_s$$

$$\therefore PG = \frac{4f_s}{f_s/4} = 16$$

$$(PG)_{dB} = 10 \log_{10}(16) = \underline{\underline{12 \text{ dB}}}$$

Application of Spread Spectrum modulation

(11)

1. The Spread spectrum modulation has the ability to resist the effect of intentional Jamming. Earlier this anti-jamming capability was used in military applications. But Now days it is used in commercial applications also.
2. Low probability of intercept is an application of Spread Spectrum in military applications. In this case, the signal spectral density is kept small such that the presence of signal is not detected easily.
3. Spread spectrum is used in mobile communication. This is because the spread spectrum signal has the ability to resist the effect of multipath fading. The BW of Spread Spectrum signal, only a small portion of the signal is in fade. The resistance of fading is important consideration in mobile communication.
4. Spread Spectrum communications are used secure. Since PN codes are used to generate Spread Spectrum signals, unwanted receivers can't recognize the Spread Spectrum signals.
5. Spread Spectrum communications are used in distance measurement. Broadband signals can be resolved in time more precisely than narrow band signals. Thus if broad band signal is transmitted, then the time delays can be measured more accurately and the distance measurement is accurate. Spread Spectrum signals are used in radars and navigation systems for distance measurement.
6. Spread Spectrum is also used in selective calling. In this, the central station communicates with no of receiving points. This is accomplished due to PN code in Spread Spectrum signal.
7. Spread Spectrum is also used in Code division multiple access (CDMA) communications. In this system many users can communicate at the same time & through the same channel.

Types of Jammers

1. The barrage noise jammer: This consists of white Gaussian noise. The noise is generated over a specific BW without the knowledge of anti-jam communication jammer.
2. Partial band noise jammer: The power of such jammer is spread over the subset of total BW. Such noise is ~~easy~~ easier to generate compared to barrage noise jammer. This jammer is effective against FH/MFSK systems.
3. Pulsed noise jammer: Such jammer transmits the wide band noise for short duration of time. This ~~is~~ means the noise is transmitted in the form of noise bursts. This jammer is effective against DS/BPSK syst.
4. Single tone jammer: This jammer generates sinusoidal waveforms of single frequency. The frequency of jamming signal lies in spread BW. This is easiest to generate.
5. Multitone jammer: It consists of multiple sinusoidal frequencies in the spread BW. Hence multitone jammer is known as partial band noise jammer.

Ex 1 A single ~~tone~~ tone jammer $j(t) = \sqrt{2J} \cos(2\pi f_c t + \theta)$ is applied to DS/BPSK syst. The N differential transmitted signal $x(t)$ is described by $x(t) = \pm \sqrt{\frac{E_b}{N}} \sum_{k=0}^{N-1} C_k \phi_k(t)$ for $0 \leq t \leq T_b$. Find $2N$ coordinates of the jammer $j(t)$.

Where $C_k \Rightarrow$ PN Sequence

$\phi_k(t) \Rightarrow$ orthogonal basis functions.

$N \Rightarrow$ length of PN sequence